LAB 12

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Regime filter
TSC reference	A4, K4, K6
Duration	15 min
Folder	labs_fullcode/Lab12-market-regimes/
Files	Lab12-market-regimes/regime_stage.py, pipeline/regime.py, Lab12-market-regimes/solution/regime.py
Prompts	Lab12_Prompts.pdf in this folder (step -> prompt, ready to copy)
Mode	Runs ONLINE with your Alpaca paper keys (.env), or OFFLINE on the bundled data and the paper-broker simulator. Paper trading only.

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Scenario

The risk team suspects the desk's signals behave differently in calm and stressed markets. It asks for an unsupervised regime model that can switch signals off in stress - fitted walk-forward so that the gate never uses information from the future.

Objective

Build pipeline/regime.py: PCA on the universe's returns, K-means regimes on backward-looking SPY features, walk-forward regime labels, and a gate that forces signals flat in the stress regime, evaluated after costs.

Before You Start

- Open a terminal in the labs_fullcode folder, activate your virtual environment and open your AI coding assistant there (set-up below). The complete, working code is already in this folder.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

How to Run the Python Scripts

Everything runs from the labs_fullcode folder, ONLINE with your Alpaca paper keys or OFFLINE on the bundled data. Do the first four rows once; after that each lab is one command.

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Step	Command / file
Open a terminal in the labs_fullcode folder	cd labs_fullcode
Create and activate a virtual environment (once)	python3 -m venv .venv source .venv/bin/activate (Windows: .venv\Scripts\activate)
Install the libraries (once)	pip install -r requirements.txt
Optional: your Alpaca paper keys (leave blank to run OFFLINE)	cp .env.example .env (Windows: copy .env.example .env) then paste APCA_API_KEY_ID and APCA_API_SECRET_KEY into .env
Run Lab 12	python3 Lab12-market-regimes/regime_stage.py (same as: python -m pipeline regime)
The complete code	pipeline/regime.py (reference copy: Lab12-market-regimes/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

Lab Steps - In Python with Your AI Assistant

AI writes the code; you own the algorithm. Paste each prompt into your AI coding assistant opened in the labs_fullcode folder (the same prompts, ready to copy, are in Lab12-market-regimes/Lab12_Prompts.pdf and Prompt_Library.html), run the stage and check the result. The complete, working code is already provided - see 'How to Run the Python Scripts' above. The Full Code Practical Performance uses this code path.

Step 1: Build PCA and the regime features (K4)

Ask your assistant for the first part of pipeline/regime.py.

AI prompt - paste into your AI coding assistant
Create labs_fullcode/pipeline/regime.py. regime_features(closes) builds backward-looking SPY features: vol_21 (annualised 21-day volatility), ret_21, ret_63, dist_sma200 (close / ta_sma(200) - 1) and corr_spy_tlt (63-day rolling correlation of SPY and TLT returns), dropping NaN rows. fit_regimes(train, k) fits StandardScaler + KMeans(n_clusters=k, n_init=20, random_state=42) and names clusters by average vol_21: 'Low vol', 'Moderate vol', 'High vol (stress)' for k=3, otherwise 'Regime i (vol rank)' with the highest named 'High vol (stress)'. In run() also fit sklearn PCA on standardised daily returns of config universe.trade (pipeline.ingest.load_closes) and print variance explained by PC1-PC4 and the loadings. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: fit_regimes returns a scaler, a fitted KMeans and a dict of names ordered from low to high volatility.

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Step 2: Add walk-forward labels and the gate (K6, A4)

Ask your assistant for the walk-forward loop, regime_gate(), run() and main().

AI prompt - paste into your AI coding assistant
In labs_fullcode/pipeline/regime.py add walk_forward_regimes(feat, k, first_year=2020): for each calendar year fit_regimes on data up to 31 December of the previous year only and label that year's days; regime_gate(target, regimes, blocked=('High vol (stress)',)) forces the target to 0 on blocked days (the regime is known at the close, so the gate acts on the next bar like any signal). run(k=3) for `python -m pipeline regime --clusters`: print [1] PCA, [2] full-sample cluster profiles (labelled hindsight, description only) with next-day SPY Sharpe and trend-filter Sharpe by regime, [3] walk-forward regime counts and the latest regime and gate state, [4] gated versus ungated CAGR, Sharpe, max drawdown, exposure and trades from 2020 for the configured strategy, trend_filter and dip_buy (backtest with config costs). Save _outputs/Lab12_regime_labels.csv and chart Lab12_regimes.png (PCA variance, SPY coloured by walk-forward regime, gated vs ungated equity). Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

Check: python -m pipeline regime --help shows --clusters.

Step 3: Run the regime stage and read the PCA (A4, K4)

Run: python -m pipeline regime --clusters 3 (same as: python3 Lab12-market-regimes/regime_stage.py).

Check: PCA on 9 assets x 2,010 days: PC1 56.8%, PC2 15.5%, PC3 10.3% (cumulative 82.5%). PC1 loads 0.33-0.43 on the equities and -0.09 on TLT: one broad equity factor; TLT and GLD are the diversifiers.

Step 4: Read the regime profiles (K4, K6)

Read section [2]. In which regime does the trend filter work, and where does it fail?

Check: Low vol 654 days (vol 12.7%), Moderate vol 786 days (vol 13.7%, stock-bond correlation -0.33), High vol (stress) 372 days (vol 30.4%). Trend-filter Sharpe 1.23 / 1.04 / -0.45 by regime.

Step 5: Check the walk-forward labels (K6)

Read section [3]. Why does the walk-forward model label far fewer stress days than the full-sample fit?

Check: 2020-2025: Low vol 1,061, Moderate vol 383, High vol (stress) 64 days; latest regime (2025-12-31) Low vol -> gate OPEN.

Step 6: Evaluate the gate after costs (A4)

Read section [4] and _outputs/Lab12_regimes.png. Should the gate go into the live pipeline?

AI prompt - paste into your AI coding assistant
Using the gated versus ungated table from python -m pipeline regime, write a short recommendation to the risk team on whether to switch the regime gate on for ema_cross, trend_filter and dip_buy. Quote Sharpe and max drawdown, mention that the gate is itself a model fitted on few stress days, and say which test from Lab 13 should decide.

Check: ema_cross 0.91 -> 0.89 and trend_filter 0.89 -> 0.80 (gate hurts); dip_buy 0.63 -> 0.72 with max drawdown -17.98% -> -12.37% (gate helps) - so the gate is signal-specific.

Step 7: Try four clusters (A4)

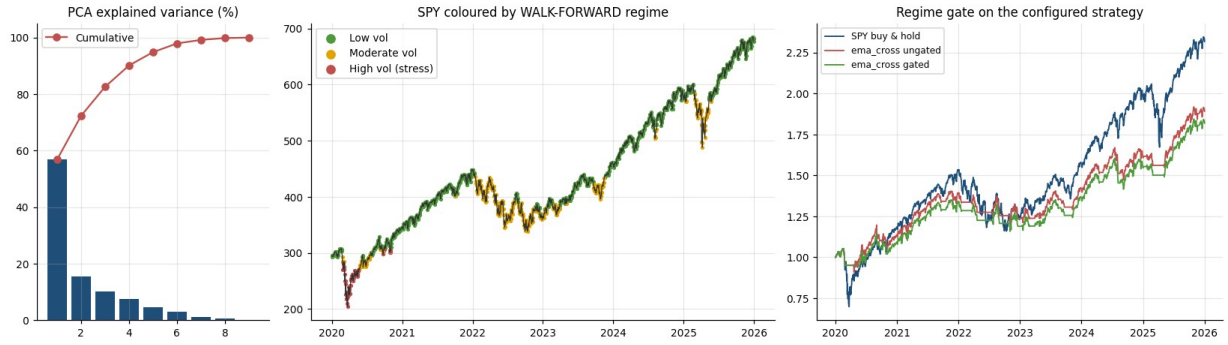
Run: python -m pipeline regime --clusters 4 and compare the profiles.

Check: Four regimes; the highest-volatility cluster has only 23 days (vol 80.8%, the March 2020 crash) - too few to trust a gate built on it.

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Expected Output (Full Code)

python -m pipeline regime --clusters 3: PC1 56.8%, PC2 15.5%, PC3 10.3% (82.5% cumulative); regimes Low vol (654 days, vol 12.7%), Moderate vol (786, 13.7%), High vol (stress) (372, 30.4%); trend-filter Sharpe 1.23 / 1.04 / -0.45 by regime; walk-forward 2020-2025: 1,061 / 383 / 64 days, latest Low vol (gate open); gate effect on Sharpe: ema_cross 0.91 -> 0.89, trend_filter 0.89 -> 0.80, dip_buy 0.63 -> 0.72; files _outputs/Lab12_regime_labels.csv and chart _outputs/Lab12_regimes.png.



Lab 12 output (labs_fullcode/_outputs/Lab12_regimes.png), offline mode on the bundled data.

Debrief

Unsupervised learning (K4) finds structure without labels: PCA compresses correlated assets into a few drivers and K-means groups days into regimes. Fitting the regimes walk-forward is part of validating the model (K6), and testing the gate signal by signal after costs (A4) shows that a filter can help one strategy and hurt another.

Troubleshooting

Symptom	What to do
Cluster names swap between runs	K-means numbers clusters arbitrarily; fit_regimes names them by volatility so labels stay meaningful.
ConvergenceWarning	Increase n_init or max_iter in KMeans; results are usually unaffected.
Walk-forward starts in 2020	The first fit needs at least 200 days of features after the 200-day warm-up.

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